

• PART - A

(1) Define meet of two Boolean matrix.

Ans The meet of two Boolean matrix A and B , denoted by $A \wedge B$, is a matrix $D = [d_{ij}]_{m \times n}$ where

$$d_{ij} = \begin{cases} 1, & \text{if } a_{ij} = 1 = b_{ij} \\ 0, & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases}$$

Symbolically, we write $A \wedge B = D$

(2) Define join of two Boolean matrix.

→ The join of two Boolean matrix of same size A and B , denoted by $A \vee B$, is a matrix $C = [c_{ij}]_{m \times n}$ where

$$c_{ij} = \begin{cases} 1, & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0, & \text{if } a_{ij} = 0 = b_{ij} \end{cases}$$

Symbolically, we write $A \vee B = C$

(3) Define Product of two boolean matrix.

→ The Boolean Product of two Boolean matrix $A = [a_{ij}]_{m \times p}$ and $B = [b_{ij}]_{p \times n}$ denoted by $A \odot B$, is the Boolean matrix

$$C = [c_{ij}]_{m \times n}, \text{ where}$$

$$c_{ij} = \begin{cases} 1, & \text{if } a_{ik} = 1 = b_{kj} \text{ for some } k, 1 \leq k \leq p \\ 0, & \text{if } a_{ik} = 0 \text{ or } b_{kj} = 0, \text{ for some } k, 1 \leq k \leq p \end{cases}$$

Symbolically, we write $A \odot B = C$

(4) Write the steps of Warshall's Algorithm.

→ Steps of Warshall's Algorithm:

Step 1: Copy all the 1's of W_{k-1} in W_k .

Step 2: Identify the positions of $s_1, s_2, \dots$, in Column k of W_{k-1} , where the entry is 1, and the positions $t_1, t_2, \dots$, in row k of W_{k-1} where the entry is 1.

Step 3: Write 1 at all the positions (s_i, t_j) of W_k , if 1 is not already written there.

5. if $A = \{a, b, c, d\}$ and R be equivalence relation on A defined by $R = \{(a, a), (a, b), (b, a), (b, b), (c, d), (d, c), (c, c), (d, d)\}$. Find equivalence classes.

Ans

$$A = \{a, b, c, d\}$$

$$R = \{(a, a), (a, b), (b, a), (b, b), (c, d), (d, c), (c, c), (d, d)\}$$

$$[a] = \{a, b\} \quad \text{equivalence class of } a.$$

$$[b] = \{b, a\} \quad \text{equivalence class of } b.$$

$$[c] = \{c, d\} \quad \text{equivalence class of } c.$$

$$[d] = \{d, c\} \quad \text{equivalence class of } d.$$

• PART - B

(1) if $A = \{1, 2, 3, 4, 5\}$ and $B = \{a, b, c\}$ and R be the relation from A to B defined by $R = \{(1, a), (1, b), (3, a), (2, c), (3, c), (2, b), (5, a), (5, c)\}$. Find R -relative set of the element 1.

→ R -Relative set of an element x :

Let R be a relation from A to B and $x \in A$, then R relative set of x , denoted by $R(x)$, is the set of those elements of B which are related to that particular element x i.e.,

$$R(x) = \{y \in B \mid (x, y) \in R\}$$

$$A = \{1, 2, 3, 4, 5\}$$

$$R = \{(1, a), (1, b), (3, a), (3, c), (2, b), (5, a), (5, c)\}$$

R - relative set of element 1:

$$R(1) = \{a, b\}$$

2.

Find the join and meet of these two matrices

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Join of A and B ($A \vee B$):

$$A \vee B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

meet of A and B ($A \wedge B$)

$$A \wedge B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3.

Find the relation R when

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Be a matrix of relation R from set

$$X = \{a, b, c\} \text{ and } Y = \{1, 2, 3, 4, 5\}.$$

Sol $\rightarrow$

$$X = \{a, b, c\}$$

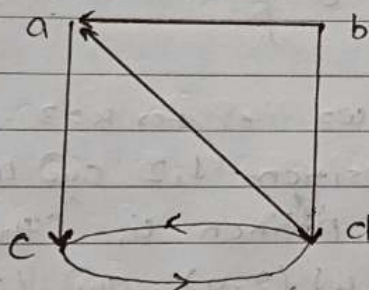
$$Y = \{1, 2, 3, 4, 5\}$$

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$R = \{(a, 2), (b, 1), (b, 3), (b, 4), (b, 5), (c, 1), (c, 2), (c, 3), (c, 4)\}$$

PART - C

- (1) Let R be the Relation with diagram show below. Find transitive closure using Warshall's Algorithm.

Sol.

Here, set $A = \{a, b, c, d\}$ so $n = |A| = 4$

Also, $R = \{(a, c), (b, a), (b, d), (d, a), (d, c), (c, d)\}$

$$\therefore M_R = W_0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

First we compute W_1 , i.e; for $k=1$. Column 1 of W_0 has 1's at 2 and 4 positions, while row 1 of W_0 has a 1 at 3 position. This W_1 1's at (2,3) and (4,3) positions along with existing 1's in W_0 .

$$\therefore W_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Next, we compute W_2 , i.e. for $k=2$. Column 2 of W_1 has no 1's so there is no new 1 to be added to W_1 to obtain W_2 .

Thus $W_2 = W_1$

$$W_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Further, we compute W_3 i.e. for $k=3$. Column 3 of W_2 has 1's at position 1, 2 and 4 while row 3 of W_2 has 1's at position 4, thus W_3 has 1's at positions (1,4), (2,4) and (4,4) along with existing 1's of W_2 .

$$W_3 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Finally, we compute W_4 i.e. for $k=4$. Column 4 of W_3 has 1's at positions 1, 2, 3 and 4, while row 4 of W_3 has 1's at position 1, 3 and 4. Thus W_4 has 1's at (1,1), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,3), (3,4), (4,1), (4,3) and (4,4) positions along with the existing 1's in W_3 .

$$W_u = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Transitive closure of $R = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$

2. (i) Consider a set $A = \{a, b, c, d, e\}$ and a relation R defined on A given by $R = \{(a,a), (a,b), (b,b), (c,c), (b,a), (d,d), (d,e), (d,f), (e,d), (e,e), (e,f), (f,d), (f,e), (f,f)\}$. Write the matrix representation M_R of the relation and hence prove that it is an equivalence relation.

solⁿ.

The set $A = \{a, b, c, d, e, f\}$

The matrix representation of given relation R is

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

- (i) Reflexive: The diagonal matrix elements of M_R are non-zero it implies that every element of the set A is related to itself hence the relation is reflexive.
- (ii), Symmetric: The matrix M_R and its transpose $(M_R)^T$ are identical hence the relation is Symmetric.
- (iii) Transitive: The matrix $(M_R)^2$ is given by

$$MR \cdot MR = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} = MR$$

So we notice that $MR \cdot MR = MR$

Thus relation is transitive

Hence, the relation is an equivalence relation.

(ii). Compute the product of the Boolean matrices

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Sol. $A \odot B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 1 & 1 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 & 1 \cdot 1 + 1 \cdot 0 + 0 \cdot 1 \\ 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 & 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 & 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 & 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 \\ 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 & 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

3. (i) Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the equivalence relation on set A defined by
- $$R = \{(1,1), (1,2), (2,3), (4,5), (1,6), (2,2), (2,4), (3,3), (4,4), (5,5), (6,6), (2,6), (5,6)\}.$$
- Find partition of A induced by R i.e., find the equivalence class.

Sol

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$R = \{(1,1), (1,2), (2,3), (4,5), (1,6), (2,2), (2,4), (3,3), (4,4), (5,5), (6,6), (2,6), (5,6)\}$$

$$[1] = \{1, 2, 6\}$$

$$[2] = \{3, 2, 4, 6\}$$

$$[3] = \{3\}$$

$$[4] = \{4, 5\}$$

$$[5] = \{5, 6\}$$

$$[6] = \{6\}$$

Partition of A induced by R is

$$\{\{1, 2, 6\}, \{3, 2, 4, 6\}, \{3\}, \{4, 5\}, \{5, 6\}, \{6\}\}.$$

- (ii) Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1,3), (3,2), (3,4), (2,4)\}$ be a relation on A . Find the transitive closure of R using Marshall's algorithm.

Sol

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1,3), (3,2), (3,4), (2,4)\}$$

Here,

$$W_0 = M_R = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{and } n = |A| = 4$$

Now, we find W_1 i.e. $k=1$. Column 1 of W_0 has no 1 at positions of Column 1 so $W_1 = W_0$

$$W_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Next, we find W_2 i.e. for $k=2$. Column 2 of W_1 has 1's at position 3 while row 2 of W_1 has 1's at position 4. Thus W_2 has 1's at (3,4) along with 1's of W_1 .

$$W_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Further, we find W_3 i.e. for $k=3$. Column 3 of W_2 has 1's at position 1 while row 3 has 1's at position 2 and 4. Thus W_3 has 1's at (1,2) and (1,4) along with 1's of W_2 .

$$W_3 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Finally, we complete W_4 i.e. $k=4$. Column 4 of W_3 has 1's at position 1, 2, 3, while row 4 does not have 1 at any position.

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So there is no 1's added to w_3 to obtain w_4 .

$$w_4 = w_3 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Transitive closure of R

$$= \{ (1,2), (1,3), (1,4), (2,4), (3,2), (3,4) \}.$$

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